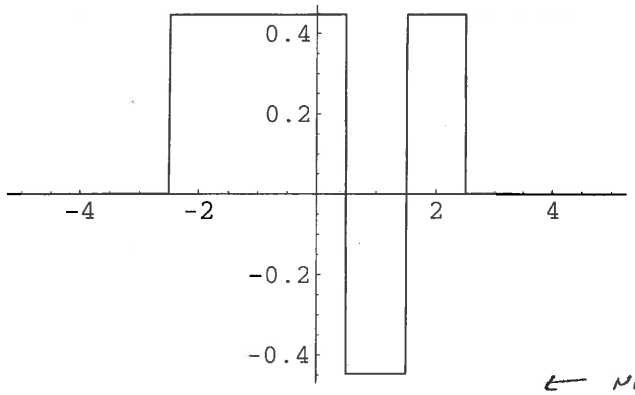
```
In[123]:= g[t_] := UnitStep[t + 1 / 2] (1 - UnitStep[t - 1 / 2])

In[124]:= Barker[t_, tp_] := (g[t / (5 tp)] - 2 g[t / tp - 1]) / Sqrt[5]

In[125]:= Plot[Barker[t, 1], {t, -5, 5}]
```



5-BIT BARKER CODE

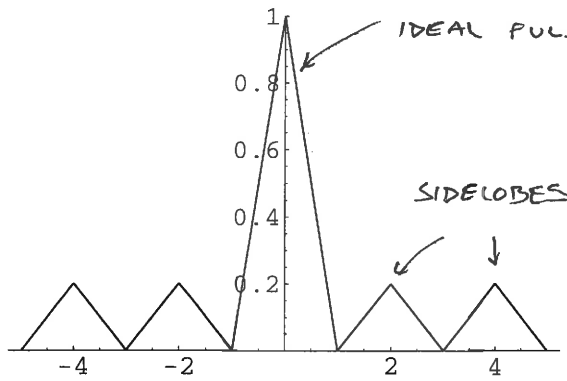
← NOTE NORMALIZATION

```
Out[125]= - Graphics -

In[126]:= Clear[chi]

In[127]:= chi[tau_, v_] := Integrate[Barker[t, 1] * Barker[t - tau, 1] * Exp[I 2 Pi v t], {t, -10, 10}]

In[128]:= Plot[Abs[chi[tau, 0.0]], {tau, -5, 5}, PlotRange → {0, 1}]
```



IDEAL PULSE COMPRESSION

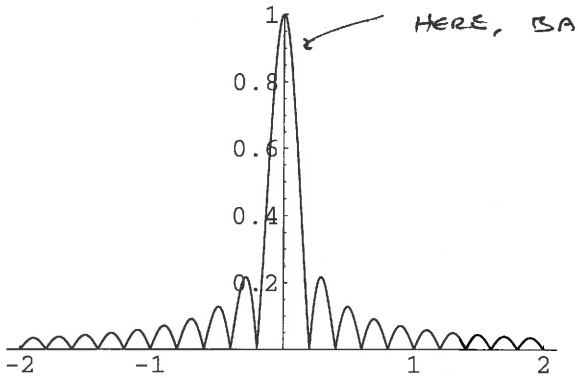
WIDTH ~ BIT WIDTH

SIDELOBES

V = 0 CUT

```
Out[128]= - Graphics -
```

In[129]:= **Plot[Abs[chi[0.0, v]], {v, -2, 2}, PlotRange → {0, 1}]**

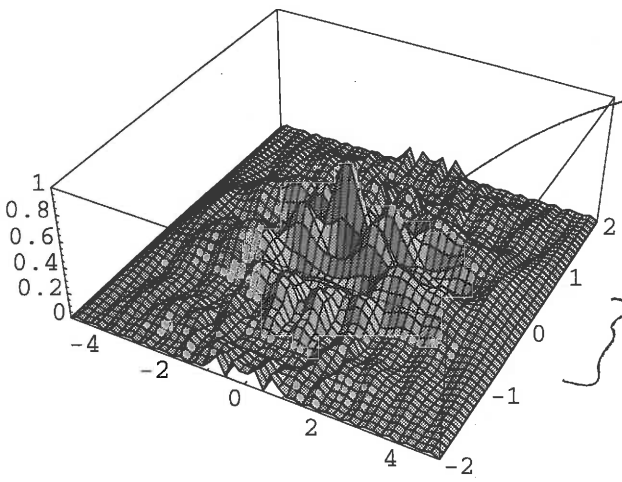


HERE, BANDWIDTH ~ $\frac{1}{\text{PULSE WIDTH}}$ (SPECIAL CASE FOR $\tau \approx 0$ ONLY)

$\tau = 0$ CUT

Out[129]= - Graphics -

In[130]:= **Plot3D[Abs[chi[tau, v]], {tau, -5, 5}, {v, -2, 2}, PlotRange → {0, 1}, PlotPoints → 50]**

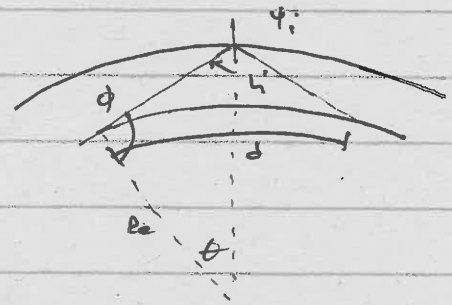


LARGE SIDELOBES REPRESENT FAILURE OF PULSE COMPRESSION FOR TARGETS WITH FINITE DOPPLER SHIFTS.

OVERALL, BANDWIDTH ~ $\frac{1}{\text{BIT WIDTH}}$

Out[130]= - SurfaceGraphics -

2



LAW OF SINES $\quad \dfrac{R_e+h'}{\sin\phi} = \dfrac{R_e}{\sin\psi_i}$

AND $\pi - \theta - \psi_i = \phi$

SO $\sin\phi = \sin(\pi-\theta-\psi_i) = \sin(\theta+\psi_i)$

HENCE $\quad \dfrac{R_e+h'}{R_e} = \dfrac{\sin\phi}{\sin\psi_i} = \dfrac{\sin(\theta+\psi_i)}{\sin\psi_i} = \dfrac{\sin\theta\cos\psi_i + \cos\theta\sin\psi_i}{\sin\psi_i}$

OR $\quad 1+\dfrac{h'}{R_e} = \sin\theta\cot\psi_i + \cos\theta \quad \div \sin\theta$

$\left(1+\dfrac{h'}{R_e} - \cos\theta\right)\csc\theta = \cot\psi_i \quad *$

3. $\theta = d/2R_e = 0.265$ rad $\Rightarrow \psi_i = 72.6°$ from $*$

MAXIMUM USABLE FREQUENCY $= f_c \sec\psi_i = 9\sqrt{N_{max}}\sec\psi_i$

$= \underline{30\ \text{MHz}}$

ELEVATION ANGLE OF RADIATION IS $\phi - \frac{\pi}{2} = \frac{\pi}{2} - \theta - \psi_i$

$= \underline{2.21°}$

AT NIGHT, MAXIMUM USABLE FREQUENCY FALLS TO $\underline{9.48\ \text{MHz}}$

31-M BAND OCCUPIES 9.2-9.7 MHz, AND THIS MUF FALLS WITHIN IT. YOU CAN ALWAYS USE A FREQUENCY LESS THAN THE MUF, ALTHOUGH IF THE FREQUENCY IS TOO LOW, ABSORPTION IN THE D REGION BECOMES PROBLEMATIC.

3. CAN USE $\frac{dx}{dy} = \sqrt{\frac{\sin^2 t_0}{n^2 - \sin^2 t_0}}$ $\qquad n^2(y) = 1 - \epsilon y^2$

$$x(y) = \int_0^y \sqrt{\frac{\sin^2 t_0}{\cos^2 t_0 - \epsilon y^2}}\, dy$$

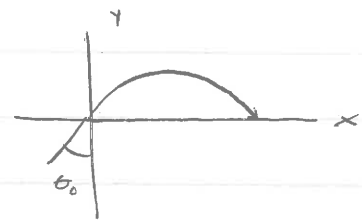
NOTE $\int \frac{dx}{\sqrt{R}} = \frac{-1}{\sqrt{c}} \sin^{-1} \frac{2cx+b}{\sqrt{-\Delta}}$ $\qquad R = a + bx + cx^2 \quad \Delta = 4ac - b^2$ $\quad$ IF $c<0$, $\Delta<0$

HERE, $a = \cos^2 t_0 \quad b = 0 \quad c = -\epsilon$

SO $x(y) = -\sin t_0 \frac{1}{\sqrt{\epsilon}} \sin^{-1} \frac{-2\epsilon y}{\sqrt{4\epsilon \cos^2 t_0}} \Big|_0^y$

$$= \sin t_0 \frac{1}{\sqrt{\epsilon}} \sin^{-1} \frac{\sqrt{\epsilon}\, y}{\cos t_0}$$

OR $y(x) = \frac{\cos t_0}{\sqrt{\epsilon}} \sin\left(\frac{\sqrt{\epsilon}\, x}{\sin t_0}\right)$



NOTE $y'(x=0) = \cot t_0$ AS EXPECTED

DEEPEST ENTRY INTO MATERIAL IS $y = \frac{\cos t_0}{\sqrt{\epsilon}}$ $\qquad$ IF $t_0 = 0$, $y_{max} = \frac{1}{\sqrt{\epsilon}}$

$y_{min} = 0$

RAY EXITS MATERIAL AT $\frac{\sqrt{\epsilon}\, x}{\sin t_0} = \pi$ OR $x = \frac{\pi}{\sqrt{\epsilon}} \sin t_0$